Theorem 2.9 Properties of Logarithm Functions

Let $x, y > 0$ be real numbers. Let $b > 0$, $b \neq 1$

Rule	Formulation
Equality Rule	$\log_b x = \log_b y \Leftrightarrow x = y$
Inequality Rules	Case $b > 1$ $\quad x < y \Rightarrow \log_b x < \log_b y$ Case $b < 1$ $\quad x < y \Rightarrow \log_b x > \log_b y$
Product Rule	$\log_b (xy) = \log_b x + \log_b y$
Quotient Rule	$\log_b \frac{x}{y} = \log_b x - \log_b y$
Exponent Rule	$\log_b x^p = p \log_b x$
Inversion Rules	$b^{\log_b x} = x, \quad \log_b b^x = x$
Special Values	$\log_b b = 1, \quad \log_b 1 = 0$